Did the 2025 Half-Cent Tick and Access-Fee Cut Tighten Spreads?
A Daily-Data Difference-in-Differences, and Why It Cannot Yet Answer the Question

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Abstract

On November 3, 2025, the SEC's amendments to Regulation NMS took effect: a $0.005 ("half-cent") minimum pricing increment for tick-constrained NMS stocks and a cut in the Rule 610 access-fee cap from $0.0030 to $0.0010 per share. We ask whether tick-constrained (treated) stocks' quoted spreads narrowed relative to wide-spread (control) stocks. Because WRDS TAQ intraday NBBO is not available on our account, we use the brief's sanctioned fallback—the CRSP daily *closing* consolidated NBBO (a Chung–Zhang-style daily spread proxy)—for 150 NMS common stocks (75 tick-constrained, 75 wide) over four weeks before and after the compliance date (5,820 stock-days). The

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difference-in-differences (DiD) estimate for the relative quoted spread is +0.73 basis points ($t = 4.2$; $t = 2.9$ double-clustered)—the *wrong* sign, driven entirely by wide-spread controls narrowing more than treated names. We do not interpret this as a treatment effect, because the design fails its own identifying checks: a placebo "event" inside the pre-period is large and significant (−0.77 bps, $t = -5.5$) and the event-study shows non-parallel pre-trends. More fundamentally, the daily close is *censored to the penny grid* for exactly the stocks the reform targets: **0.0%** of treated stock-days show a sub-penny closing spread either before or after the reform, and the share pinned at exactly \$0.01 *rose* from 81% to 89%. Both treated and control dollar spreads fell from October to November (treated −10.7%, control −22.6%; both $p < 0.001$), a market-wide move the daily design cannot separate from the reform. Four extensions turn this from "we found nothing" into an informative, design-based null. *Heterogeneity:* no observable subgroup (by volatility, volume, listing venue, or industry) shows the predicted narrowing—the wrong sign is uniform. *Overlap:* treated and control stocks are balanced on volatility and relative spread but differ by a normalized 2.4 standard deviations in log price and 1.1 in log volume, so the wide-spread controls are not a counterfactual—motivating a price-matched control. *Randomization inference* (500 permutations of treatment) places the actual estimate far outside the permutation distribution ($p < 0.002$), confirming it reflects the systematic price/liquidity structure of *which* stocks are treated, not the reform. *Power:* at 80% power the design detects a 0.49 bps narrowing, below the ≈1.0 bps a half-cent move would produce at the median treated price—so the effect would be detectable *if* the daily close could take half-cent values, which the censoring table shows it never does. We report a transparent, measurement-limited **null** and specify the TAQ intraday test required to answer the question.

JEL: G14, G18, G12, G28. **Keywords:** tick size, minimum pricing increment, access fees, bid–ask spread, Regulation NMS, difference-in-differences, randomization inference, statistical power, market microstructure.

1 Introduction

The size of the "tick"—the minimum increment in which securities may be quoted—is one of the oldest and most consequential design parameters of an equity market. Since decimalization in 2001, U.S. NMS stocks priced at or above \$1 have quoted in whole pennies. For actively traded, lower-priced stocks the penny is often *binding*: the quoted bid–ask spread sits at exactly \$0.01 for most of the day because market makers would quote tighter if the rules let them [Harris, 1994, O'Hara et al., 2019]. These stocks are called *tick-constrained*. On September 18, 2024 the Securities and Exchange Commission adopted amendments to Regulation NMS that, effective November 3, 2025, create a *variable* minimum increment: NMS stocks whose time-weighted average quoted spread falls at or below \$0.015 may quote in \$0.005 half-cents, while all other stocks remain on the penny [SEC, 2024]. The same rulemaking cut the Rule 610 access-fee cap from \$0.0030 to \$0.0010 per share. The stated goal is tighter spreads and lower trading costs for tick-constrained stocks.

This paper asks the natural first question: *did quoted spreads actually narrow for tick-constrained stocks after November 3, 2025, relative to stocks the reform did not bind?* We set up a difference-in-differences (DiD) comparison of 150 NMS common stocks—75 tick-constrained (treated) and 75 wide-spread (control)—over the four weeks either side of the compliance date. Our headline is not the answer we hoped to give, and the value of the paper is in explaining *why* honestly rather than manufacturing a result.

Preview of results. First, the DiD point estimate for the relative quoted spread is +0.73 basis points (bps), statistically significant ($t = 4.2$ clustered by stock; $t = 2.9$ double-clustered by stock and day), and stable across specifications with stock and day fixed effects and price/volume controls (Table 3). The positive sign means treated stocks' spreads did *not* narrow relative to controls—if anything they widened in relative terms. Second, and decisively, this DiD does not identify a causal effect: a placebo test that assigns a fake "event" inside the pre-period yields a large, significant coefficient of -0.77 bps ($t = -5.5$; Table 6), and the event-study (Figure 2) shows the treated–control gap wandering significantly *before* November 3. The parallel-trends assumption fails, so the DiD is uninformative about causation. Third, we trace the failure to two forces. (i) The control group (high-priced mega-caps, median price \$262) is not a valid counterfactual for treated stocks (median price \$44): their prices and spreads move on different macro betas. (ii) More fundamentally, the daily *closing* quote is censored to the penny grid for the treated stocks: **0.0%** of treated stock-days show a sub-penny closing spread either before or after the reform (Table 8, Figure 3), and the share pinned at exactly one cent actually *rose* from 81.3% to 89.1%. The half-cent margin on which the rule operates is an *intraday* phenomenon that a once-a-day closing snapshot cannot resolve. Fourth, as description (not causal inference), both groups' dollar spreads fell from October to November—treated by -10.7% ($1.33 \to 1.19$ cents, paired $t = -3.6$) and control by -22.6% ($20.3 \to 15.7$ cents, paired $t = -4.6$; Table 7)—a broad late-2025 tightening the daily design cannot attribute to the reform.

Beyond "we found nothing." A paper that reported a wrong-signed coefficient and stopped would be uninformative. We take four further steps, each using only observable data and each a design-based statement a referee can check. First, *heterogeneity* (Table 9): we interact the DiD with four pre-rule moderators that vary within both arms—September return volatility, dollar volume, Nasdaq listing, and financial industry—and find the DiD is positive and significant in *every* subgroup with uniformly insignificant interactions; no observable stratum shows the narrowing H1 predicts, so the null is not an average that hides a real effect somewhere. Second, an *overlap diagnostic* (Table 10): normalized differences show the two groups are balanced on volatility (0.06) and relative spread (0.11) but wildly imbalanced on log price (-2.42) and log dollar volume (-1.15)—roughly ten times the 0.25 rule-of-thumb—so the wide-spread controls are structurally not a counterfactual, and only 17 control stocks even fall inside the treated price range. This is what a valid design must fix, and it motivates the student's price-and-volatility-matched control (deliberately reserved, Section 9). Third, we show the wrong-signed DiD survives every reasonable redefinition of the observable screen (Table 11) and, under *randomization inference* with 500 permutations of the treatment label (Table 12, Figure 4), lies far outside the permutation distribution ($p < 0.002$)—which, because random assignment ignores price and liquidity, locates the estimate in the systematic price structure of *which* stocks are treated, not in the reform. Fourth, a *minimum-detectable-effect* calculation (Table 13) shows the null is informative rather than underpowered: at 80% power we could have detected a 0.49 bps relative narrowing, below the ≈ 1.0 bps a half-cent quote would produce at the median treated price—so the effect *would* be visible if the daily close could represent it, and the censoring result explains why it cannot.

Interpretation. We therefore report a *measurement-limited null*: using the best daily proxy available to us, we find no evidence that the half-cent tick tightened tick-constrained spreads relative to controls, and we show precisely why the daily proxy is structurally unable to detect the reform's effect. This is an honest and, we think, instructive outcome for a student project: it demonstrates the discipline of pre-trend and placebo testing, the difference between a *dollar* and a *relative* spread, and the gap between a daily liquidity proxy and the intraday reality it approximates [Holden and Jacobsen, 2014, Chung and Zhang, 2014]. The definitive test—a time-weighted intraday quoted spread and a trade-based effective spread from TAQ, on hand-verified tick-constrained eligibility—is laid out as the project's next phase (see `STUDENT_TASKS.md`).

Contribution. We contribute to three literatures. Relative to the classic tick-size literature [Harris, 1994, Angel, 1997, Goldstein and Kavajecz, 2000, Bessembinder, 2003], which studies moves *up* the price grid (eighths → sixteenths → pennies), we study the first move *below* the penny for U.S. equities. Relative to the modern tick and access-fee literature [O'Hara et al., 2019, Comerton-Forde et al., 2019, Albuquerque et al., 2020] built on the 2016 Tick Size Pilot, we examine the opposite policy—a tick *reduction*—in real time. Relative to the liquidity-measurement literature [Corwin and Schultz, 2012, Chung and Zhang, 2014, Holden and Jacobsen, 2014], we provide a concrete cautionary case in which a widely-used daily spread proxy is, by construction, blind to the very margin a reform changes. We do not claim to estimate the reform's effect; we claim to have bounded what daily data can and cannot say about it.

The paper proceeds as follows. Section 2 describes the rule. Section 3 states hypotheses. Section 4 places the paper in the literature. Section 5 describes data, the sample, and the design-based checks. Section 6 presents the DiD, event-study, and placebo results. Section 7 presents robustness and the mechanism (penny-grid censoring). Section 8 presents the four design-based extensions—heterogeneity, overlap, randomization inference, and power. Section 9 concludes.

2 Institutional background

Regulation NMS Rule 612 (the "sub-penny rule") has, since 2005, prohibited displaying, ranking, or accepting quotations in NMS stocks priced \$1 or more in increments smaller than \$0.01. The 2024 amendments replace the single penny increment with two tiers. A stock is assigned the \$0.005 increment for an evaluation period if its *time-weighted average quoted spread* (TWAQS) over a prior measurement period is at or below \$0.015; otherwise it keeps the \$0.01 increment. Assignments are recomputed periodically, so a stock's tier can change over time. In parallel, the Rule 610 cap on access fees (the fee an exchange may charge to take displayed liquidity) falls from \$0.0030 to \$0.0010 per share for stocks priced \$1 or more. The two changes are complementary: a smaller tick makes displayed liquidity cheaper to *provide*, and a smaller access fee makes it cheaper to *take*. The SEC's stated first compliance date for these provisions is November 3, 2025, which we use as the event date throughout.

Two features matter for identification. First, eligibility is defined *on the quoted spread* *itself*, so treatment is mechanically related to the outcome; we address this by assigning treatment from a screening window (September 2025) that is separated in time from the outcome windows. Second, the exact per-tranche phase-in schedule and the precise list of eligible symbols are set by the exchanges in a sequence of notices; hand-verifying those dates and the per-stock eligibility against the primary SEC and exchange documents is a central manual task for the student (Section 9) and is *not* performed by the automated pipeline here.

3 Hypotheses

Standard microstructure theory predicts that relaxing a binding tick lets competitive liquidity suppliers post tighter quotes, reducing quoted and effective spreads for constrained stocks, while leaving unconstrained stocks unaffected [Harris, 1994, Foucault et al., 2013].

> **H1 (quoted spread).** After November 3, 2025, tick-constrained (treated) stocks' quoted spreads fall relative to wide-spread (control) stocks: the DiD coefficient is *negative.*
>
> **H2 (effective spread).** The same holds for effective spreads, if the tighter quotes are actually traded against.
>
> **H3 (measurement resolution).** Because the reform's margin ($\$0.01 \rightarrow \$0.005$) is sub-penny and intraday, a daily *closing*-quote proxy may be unable to detect H1–H2 even if they hold; the closing quote of a constrained stock is pinned to the penny grid.

Our results reject H1 in sign (the DiD is positive) but, through the placebo and event-study, show that rejection is not credible as a causal statement; the evidence instead supports H3—the daily proxy is uninformative about the reform's true margin.

4 Related literature

This paper connects three strands. The first is the microstructure of tick size. Theory holds that a binding minimum increment inflates quoted spreads and rations price competition among liquidity suppliers, so that relaxing it should tighten spreads for constrained stocks while leaving unconstrained stocks unaffected [Harris, 1994, Foucault et al., 2013]. The empirical literature has, until now, almost entirely studied moves *up* the grid—the NYSE's shift from eighths to sixteenths [Goldstein and Kavajecz, 2000], decimalization [Bessembinder, 2003], and the 2016–2018 Tick Size Pilot that *widened* the tick for small-cap stocks [O'Hara et al., 2019, Comerton-Forde et al., 2019, Albuquerque et al., 2020]. Findings are mixed but broadly consistent with a tick that binds: wider ticks widen spreads and can shift volume across venues, while relative tick size shapes the trading environment [O'Hara et al., 2019]. We study the first move *below* the penny for U.S. equities and, unusually, in real time around the compliance date; our contribution is less a new estimate than a precise account of why the standard daily liquidity proxy cannot deliver one.

The second strand is liquidity measurement from low-frequency data. Because intraday TAQ is costly and, as here, often unavailable, a large literature builds daily spread proxies [Corwin and Schultz, 2012, Chung and Zhang, 2014] and benchmarks them against intraday truth [Holden and Jacobsen, 2014]. The consensus is that daily proxies track intraday spreads *on average* but degrade exactly where the microstructure is finest—for low-priced, penny-pinned stocks whose quoted spread is a discrete grid variable. Our penny-grid censoring result is a sharp, policy-relevant instance: the daily closing quote of a tick-constrained stock is a censored measurement that is, by construction, blind to the half-cent margin the reform changes.

The third strand is the credibility of difference-in-differences and the interpretation of null results. Recent methodological work stresses that pre-trend tests have limited power and can license false confidence [Roth, 2022], that event-study/DiD estimands are fragile under treatment-effect heterogeneity [Callaway and Sant'Anna, 2021], and that inference with a moderate number of clusters benefits from design-based methods such as randomization inference [Young, 2019]. Methodologists further urge that an insignificant estimate be read through its confidence interval and power rather than as proof of no effect [Abadie, 2020], and that covariate overlap be assessed with normalized differences before any comparison is trusted [Imbens and Rubin, 2015]. We use these tools diagnostically: a wrong-signed DiD that is statistically "significant" fails a placebo, is uniform across subgroups, rests on groups with no price overlap, and—read correctly—is confirmed by randomization inference to be a price-structure artifact rather than a treatment effect. Reporting the design's minimum detectable effect then converts the null into a bound rather than a shrug.

5 Data and sample

Source and the TAQ constraint. The ideal data are intraday NBBO quotes from WRDS TAQ, from which one computes a time-weighted quoted spread and a trade-based effective spread. On our WRDS account only the TAQ *sample* libraries are subscribed, and they cover a handful of 2008, 2009, and 2021 days—there is no 2025 TAQ. Following the project brief, we therefore use the sanctioned fallback: the CRSP daily stock file (CIZ format, `crsp.dsf_v2`), which records each security's daily *closing* consolidated NBBO (`dlybid`, `dlyask`) and covers 2025 through year-end. We define the dollar quoted spread as ask − bid and the relative quoted spread as (ask − bid)/midpoint, the daily proxy analyzed by Chung and Zhang [2014]. We also form a crude closing *effective*-spread proxy, $2\,|\mathrm{close} - \mathrm{mid}|/\mathrm{mid}$, and report it only as a caveated robustness outcome; a true effective spread requires trade prices from TAQ.

Sample construction. From all NMS common stocks (CRSP share type NS, security type EQTY/COM, issuer CORP or REIT) on NYSE, Nasdaq, or NYSE American with price $\geq$ \$1 and a valid closing quote, we compute each stock's *screening* statistics over September 2–30, 2025—a window deliberately separated from the outcome windows so that treatment assignment does not mechanically coincide with the measured outcome. A stock is **treated** (tick-constrained) if its September mean dollar quoted spread is $\leq$ \$0.015, the SEC's own TWAQS cutoff, and **control** if it is $\geq$ \$0.020 (clearly not penny-pinned). To keep the sample small and economically comparable in activity, we take the 75 most actively traded (by dollar volume) names in each arm, yielding 150 stocks. The daily panel runs August 25–December 19, 2025; the DiD analysis window is the four weeks before (October 6–31) and after (November 3–28) the event, giving 5,820 stock-days.

Summary and balance. Table 1 reports pre-period means. By construction the groups differ sharply on the assignment variable—the dollar quoted spread is 1.33 cents for treated versus 20.31 cents for control—and, because tick-constrained stocks are systematically lower-priced, on price (ln price 3.80 vs. 5.75, $t = -68.6$). Notably the *relative* spread is only modestly different (3.55 vs. 4.07 bps): a penny on a \$44 stock and eight cents on a \$262 stock are similar in proportional terms. This large level difference in price is the source of the identification problem documented below: the two groups are not natural counterfactuals for one another.

Design-based checks. Beyond the parametric DiD we run four checks that do not lean on the parallel-trends null, all using only observable screening and panel data. (1) *Heterogeneity*: we interact the DiD with pre-rule moderators that vary within both arms (September volatility, dollar volume, listing venue, industry), excluding price because it is near-collinear with treatment, to test whether any observable subgroup shows the predicted narrowing. (2) *Overlap*: we compute Imbens–Rubin normalized differences $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$ for each screening covariate and count how many control stocks fall inside the treated price range, quantifying whether a common-support comparison is even possible. (3) *Randomization inference*: we permute the time-invariant treatment label across the 150 stocks 500 times, rebuild the DiD, and compare the actual coefficient to the permutation distribution. (4) *Power*: we compute the minimum detectable effect at 80% power from the preferred specification's standard error. We do *not* construct the price-and-volatility-matched control group and re-estimate on it: that hand-curated match, drawn from the full NMS universe rather than only the wide-spread arm, is a reserved manual contribution for the student (Section 9), and our overlap diagnostic is designed to motivate it, not to preempt it.

6 Results

The 2×2. Table 2 shows the raw pre/post means. The relative quoted spread of *control* stocks fell from 4.07 to 3.37 bps (−0.70), while *treated* stocks were essentially flat (3.55 to 3.59, +0.03). The implied DiD is +0.73 bps. In dollar terms controls fell 4.57 cents (from 20.31) while treated fell only 0.14 cents (from 1.33), a DiD of +4.42 cents. Both DiDs are positive: the wide-spread controls narrowed more than the tick-constrained treated group.

Regression DiD. Table 3 estimates

$$\text{RelSpread}_{it} = \alpha_i + \delta_t + \beta\,(\text{Treated}_i \times \text{Post}_t) + \gamma' X_{it} + \varepsilon_{it},$$

with stock (α_i) and day (δ_t) fixed effects, controls X for ln price and ln dollar volume, and standard errors clustered by stock. The DiD β is +0.73 bps and highly stable across the five specifications (+0.73 to +0.74), with t-statistics of 4.2 clustered by stock and 2.9 double-clustered by stock and day. The preferred specification (4) has a 95% confidence interval of $[+0.39, +1.07]$ bps and a minimum detectable effect of about 0.49 bps at 80% power—so we could have detected a narrowing of half a basis point had one existed in this proxy. The estimate is the wrong sign for H1.

Event-study and pre-trends. Table 5 and Figure 2 plot weekly DiD coefficients relative to the week before the event. If the design were valid, the pre-event coefficients would be flat near zero and any effect would appear as a downward step at week 0. Instead, weeks −6, −5, and −4 are significantly *positive* (0.88, 0.82, 0.58 bps; $t = 3.8, 3.2, 2.7$), the gap closes by weeks -2 and -1, and then re-widens after the event. The treated–control gap is wandering for reasons unrelated to November 3.

Placebo. Table 6 makes the point sharply. We drop the true post-period entirely and assign a *fake* event date of October 6 inside the pre-period. A valid design would return zero. Instead the fake-DiD is -0.77 bps ($t = -5.5$)—larger in magnitude than, and opposite in sign to, the true-period estimate. The parallel-trends assumption is violated, and the DiD cannot be read as causal.

7 Robustness and mechanism

Alternative outcomes. Table 4 re-estimates the DiD on the dollar quoted spread and on the closing effective-spread proxy. The dollar-spread DiD is +4.4 cents ($t \approx 4.5$)—again positive, again driven by controls. The effective-spread proxy DiD is small and insignificant (+0.6 to +0.7 bps, $t = 1.5$–1.8). None supports H1–H2.

Why the daily proxy is blind: penny-grid censoring. The decisive evidence is in Table 8 and Figure 3. For the treated stocks the closing dollar spread is **never** sub-penny: 0.0% of treated stock-days show a spread below one cent, both before *and* after the reform. The share pinned at exactly one cent *rose* from 81.3% to 89.1%. A closing quote that is always $\geq \$0.01$ cannot reveal a half-cent quote, so the reform's margin is invisible in these data by construction. The modest within-treated dollar narrowing we do see (-10.7%, Table 7) is stocks moving from two cents to one cent (penny compression), not one cent to a half-cent. This is a clean illustration of H3 and of the proxy limits emphasized by Holden and Jacobsen [2014].

A market-wide tightening, not a treatment. Table 7 reports paired within-group changes. Both groups' dollar spreads fell significantly from October to November—treated

-10.7% (paired $t = -3.6$) and control -22.6% (paired $t = -4.6$). Whatever narrowed spreads in late 2025 (rising prices, lower volatility, or the access-fee cut acting on *all* stocks) hit the unconstrained controls harder than the pinned treated group. A DiD that uses these controls as the counterfactual therefore attributes *negative* net tightening to the reform—a mechanical artifact of an invalid control group, not evidence that the reform backfired.

What we do not claim. We do not claim the reform failed. We claim that daily closing quotes cannot test it: the treated outcome is censored at the penny, the control group is not comparable, and the parallel-trends and placebo checks fail. The sign and significance of our DiD are real numbers from real data, but they are not a treatment effect.

8 Design-based extensions

The naive DiD is statistically significant, wrong-signed, and non-causal. This section makes that diagnosis rigorous and quantitative with four design-based checks, each on observable data.

No subgroup shows the effect. If the reform tightened spreads for *some* tick-constrained stocks and the pooled test merely averaged it away, the narrowing should surface in an observable subgroup. Table 9 interacts the DiD with four pre-rule moderators that vary within both arms—above-median September return volatility, above-median dollar volume, a Nasdaq-listing indicator, and a financial-industry indicator—reporting the base DiD (low-moderator group) and the DiD$\times$moderator interaction for both the relative and the dollar spread. Price is deliberately excluded because it is near-collinear with treatment. The result is stark: the DiD is *positive and significant in every subgroup* (relative-spread base coefficients $+0.73$ to $+1.00$ bps, all $t > 2.6$), and every interaction is statistically insignificant. No observable stratum exhibits the negative sign H1 predicts; the wrong-signed estimate is a uniform feature of the sample, not a diluted true effect hiding in a subgroup.

The control group is not a counterfactual. Table 10 reports normalized differences between treated and control on the September screening covariates. The groups are well balanced on the two dimensions that plausibly drive spread *dynamics*— return volatility (+0.06) and the relative spread itself (+0.11), both far inside the 0.25 rule-of-thumb—but are grossly imbalanced on price (ln price normalized difference −2.42) and dollar volume (−1.15), an order of magnitude beyond that threshold. Only 17 of the 75 control stocks fall anywhere inside the treated price range of \$5.9–\$177.8. A difference-in-differences that compares \$44 tick-constrained names to \$262 mega-caps is not estimating a treatment effect; it is differencing two populations that share almost no common support on price and liquidity. This is precisely the imbalance a matched control must repair, and building that match—on price and volatility, drawn from the full universe—is the student's reserved task, motivated but deliberately not performed here.

The wrong sign survives every screen redefinition. One might worry that the result depends on where we drew the \$0.015 treated cutoff or the \$0.02 control cutoff. Table 11 re-estimates the two-way fixed-effects DiD on subsets of the same 150 stocks under a tighter treated screen (September dollar spread $\leq$ \$0.012), a narrower control screen ($\geq$ \$0.03), a donut that drops names near the \$0.015 boundary, and a sample that excludes mega-caps (screen price > \$500). The DiD stays positive and significant throughout, ranging from +0.39 bps (t = 2.6, dropping mega-caps) to +1.00 bps (t = 4.8, donut). Even removing the highest-priced controls—the most obvious source of non-comparability—leaves a significant wrong-signed estimate. The pathology cannot be screened away; it requires the price-matched control the overlap diagnostic calls for.

Randomization inference locates the estimate in the price structure. Table 12 and Figure 4 report design-based inference from 500 permutations of the treatment label across the 150 stocks. For the relative quoted spread the actual DiD (+0.73 bps) lies far outside the permutation 95% interval ($[-0.37, +0.36]$), randomization $p < 0.002$; the dollar-spread estimate is likewise extreme (+4.36 cents versus $[-1.99, +2.03]$, $p < 0.002$). This does *not* rescue a causal reading. Because random assignment ignores price and liquidity, an estimate this extreme means the treated–control spread gap is a *systematic* function of *which* stocks are labeled treated—their price and volume—exactly the artifact the placebo, the pre-trends, and the overlap diagnostic identify. Randomization inference thus corroborates both halves of the story: the estimate is not sampling noise, and it is not the reform.

The null is informative, not underpowered. Finally, Table 13 reports the minimum detectable effect (MDE) at 80% power, $\text{MDE} = (z_{0.975} + z_{0.80})\,\widehat{\text{SE}} \approx 2.80\,\widehat{\text{SE}}$, for each outcome. For the relative quoted spread the MDE is 0.49 bps—13.8% of the pre-period treated mean—so a narrowing of half a basis point would have been detected had it appeared in the daily close. The economic benchmark makes the point decisive: a move from a one-cent to a half-cent quoted spread at the median treated midpoint of $48.90 is a 1.02 bps relative narrowing, *twice* the MDE. The design is therefore easily powerful enough to see the reform's mechanical margin—*if* the daily closing quote could represent a half-cent spread. The censoring result (Table 8) shows it never does: 0.0% of treated stock-days are sub-penny. The null is not the symptom of an underpowered test; it is the signature of a censored measurement. (The dollar-spread MDE is large relative to its small treated mean, so we read that noisy margin cautiously.)

9 Conclusion

Using the CRSP daily closing NBBO for 150 NMS stocks around the November 3, 2025 half-cent tick and access-fee reform, we find no credible evidence that tick-constrained stocks' spreads narrowed relative to wide-spread controls. The naive DiD is the wrong sign (+0.73 bps, $t = 4.2$), but it fails a placebo (−0.77 bps, $t = -5.5$) and shows non-parallel pre-trends, and—most importantly—the daily close is censored to the penny grid for exactly the stocks the reform targets (0.0% of treated stock-days sub-penny). Both groups' dollar spreads fell market-wide (treated −10.7%, control −22.6%), which the daily design cannot separate from the reform. The honest conclusion is that this question cannot be answered with daily data.

The path to an answer is specific and is the project's next phase. First, obtain TAQ intraday NBBO (or a vendor daily-spread product built from it) and compute a time-weighted quoted spread and a trade-based effective spread—the measures on which the reform actually operates. Second, *hand-verify tick-constrained eligibility per stock* against the SEC adopting release and the exchange phase-in notices, and log the exact per-tranche compliance dates, since the automated $0.015 screen is only an approximation of the regulatory TWAQS assignment. Third, sharpen identification with a control group matched on price and volatility, or a regression-discontinuity design around the $0.015 cutoff, so that treated and control stocks share a counterfactual. With those three pieces the same DiD skeleton built here becomes a credible test. Until then, the responsible finding is a transparent null with a documented reason.

Limitations. (i) Daily closing quotes, not intraday—the central limitation. (ii) A small, deliberately illustrative sample of 150 liquid names, not the full NMS universe. (iii) Treatment assigned by an automated spread screen, pending the student's hand-verification of true eligibility. (iv) The four-week windows are short by design (to isolate the event) and cannot speak to longer-run adjustment.

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Tables

Table 1: Pre-period covariate balance, treated vs. control

Variable (pre-period mean)	Treated (tick-constr.)	Control (wide)	Diff. (T−C)	t
Relative quoted spread (bps)	3.555	4.068	-0.513	-4.38
Dollar quoted spread (cents)	1.334	20.312	-18.978	-19.37
ln(price)	3.800	5.752	-1.952	-68.65
ln($ volume)	20.561	21.312	-0.751	-26.32
ln(#trades)	11.991	11.943	0.048	1.04

Notes: Means over the pre-period (Oct 6–31, 2025) across stock-days. Treated = tick-constrained (September mean dollar quoted spread ≤ \$0.015); control = wide-spread (≥ \$0.020). Column "t" is a Welch two-sample t on stock-day observations (illustrative; the DiD uses stock-clustered SE). 75 treated, 75 control.

Table 2: Difference-in-differences 2×2 means

	Relative spread (bps)			Dollar spread (cents)		
	Pre	Post	Δ	Pre	Post	Δ
Control (wide)	4.068	3.365	-0.702	20.312	15.746	-4.566
Treated (tick-constrained)	3.555	3.585	0.030	1.334	1.192	-0.143
DiD			0.733			4.424

Notes: Cell means of the (winsorized 1/99) relative quoted spread (bps) and dollar quoted spread (cents). Pre = Oct 6–31; Post = Nov 3–28, 2025. The bottom row is the DiD (treated change − control change).

Table 3: Main DiD: relative quoted spread (bps)

Specification	Treated×Post	t	SE	N	R^2
(1) 2x2, no FE	0.733***	(4.22)	0.174	5820	0.007
(2) + Stock FE	0.739***	(4.19)	0.176	5820	0.564
(3) TWFE	0.737***	(4.16)	0.177	5820	0.581
(4) TWFE + controls	0.732***	(4.18)	0.175	5820	0.582
(3d) TWFE, cluster stock+day	0.737***	(2.93)	0.251	5820	0.581

Notes: DV = relative quoted spread (bps, winsorized 1/99). Coefficient on Treated×Post. Specs (2)–(4) add stock FE, day FE, and ln price/ln $volume controls. SE clustered by stock; row (3d) double-clusters by stock and day. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A negative coefficient would support H1; the estimate is positive.

Table 4: Alternative outcomes

Outcome / specification	Treated×Post	t	SE	N	R^2
Dollar spread (cents), TWFE	4.448***	(4.38)	1.016	5820	0.823
Dollar spread (cents), +controls	4.360***	(4.57)	0.954	5820	0.825
Eff-spread proxy (bps), TWFE	0.604	(1.50)	0.402	5820	0.346
Eff-spread proxy (bps), +ctrl	0.718*	(1.82)	0.395	5820	0.349

Notes: DiD (Treated×Post) for the dollar quoted spread (cents) and the closing effective-spread proxy (bps), TWFE with/without controls, SE clustered by stock. The effective-spread proxy $2|\text{close} - \text{mid}|/\text{mid}$ is a crude daily measure and is reported only for completeness.

Table 5: Event-study: weekly DiD coefficients

Event week	Treated×week coef. (bps)	t
-6	0.878***	(3.78)
-5	0.821***	(3.22)
-4	0.583***	(2.69)
-3	0.195	(0.80)
-2	0.173	(0.75)
-1	0.000 (base)	
0	0.543**	(2.28)
1	0.736***	(2.90)
2	1.598***	(5.88)
3	1.029***	(3.79)
4	1.427***	(5.08)
5	1.071***	(4.87)
6	1.352***	(4.77)

Notes: Coefficients on Treated×(event-week) from a TWFE regression of the relative quoted spread (bps), base week = -1. Week 0 contains Nov 3, 2025. Significant pre-event coefficients (weeks $-6, -5, -4$) indicate non-parallel trends. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Placebo: fake event inside the pre-period

Test	Treated×Post$_{\text{fake}}$	t	SE	N
Placebo fake-event 2025-10-06 (pre-period only)	-0.772***	(-5.47)	0.141	5967

Notes: The true post-period is dropped; a fake event of Oct 6, 2025 is assigned within the pre-period (Sep 8–Oct 31). A valid design would return zero; the large significant coefficient shows the parallel-trends assumption fails.

Table 7: Within-group pre/post changes (description)

Group	Measure	Pre	Post	Δ	%Δ	paired t
Treated (tick-constrained)	Dollar spread (cents)	1.335	1.192	-0.142	-10.7%	-3.62***
Treated (tick-constrained)	Relative spread (bps)	3.548	3.581	0.033	0.9%	0.38
Control (wide)	Dollar spread (cents)	20.293	15.699	-4.593	-22.6%	-4.60***
Control (wide)	Relative spread (bps)	4.066	3.360	-0.706	-17.4%	-4.68***

Notes: Stock-level pre→post changes, averaged within group, with a paired t-test across the 75 stocks in each arm. Both groups' dollar spreads fall; the control (unconstrained) group falls more.

Table 8: Penny-grid censoring of the daily closing quote (mechanism)

Group	Period	Stock-days	Share =1.0c	Share <1.0c	Mean $ spread (c)
Treated (tick-constrained)	Pre	1484	81.3%	0.0%	1.334
Treated (tick-constrained)	Post	1419	89.1%	0.0%	1.192
Control (wide)	Pre	1498	9.5%	0.0%	29.977
Control (wide)	Post	1419	20.7%	0.0%	17.729

Notes: Share of stock-days whose closing dollar spread equals exactly $0.01 and whose spread is strictly below $0.01 (sub-penny). Treated stocks are never sub-penny at the close, before or after the reform, so the $0.005 margin is not observable in daily data. “Mean $ spread” is the raw (unwinsorized) closing dollar spread in cents.

Table 9: Heterogeneity of the DiD by observable pre-rule moderators

	Relative spread (bps)		Dollar spread (cents)	
Moderator M (pre-rule)	DiD	DiD$\times M$	DiD	DiD$\times M$
High Sept. volatility	+0.929***	-0.310	+5.596***	-2.058
High dollar volume	+1.005***	-0.457	+6.084***	-2.404
Nasdaq-listed	+0.900***	-0.227	+4.398***	+0.219
Financial industry	+0.729***	-0.073	+4.515***	-1.756

Notes: Each row adds an interaction DiD$\times M$ (and Post $\times$ M) to the two-way fixed-effects specification, where M is a pre-rule moderator that varies within both arms: above-median September return volatility, above-median dollar volume, a Nasdaq-listing indicator, and a financial-industry indicator. Columns report the base DiD (low-M group) and the DiD$\times M$ interaction for the relative and dollar quoted spread. SE clustered by stock. Price is deliberately excluded as a moderator (near-collinear with treatment). No subgroup shows the negative (narrowing) sign H1 predicts. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Covariate overlap: normalized differences (treated vs. control)

Covariate (Sept. screen)	Treated mean	Control mean	Norm. diff.
Price ($)	57.521	447.768	-0.81
ln price	3.745	5.711	-2.42
Sept. return vol.	0.028	0.026	+0.06
ln dollar volume	20.651	21.421	-1.15
Rel. spread (screen)	0.000	0.000	+0.11

Notes: September screening-window means and the Imbens–Rubin normalized difference $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$ for each covariate. Values above ≈ 0.25 in absolute terms indicate serious imbalance. The groups are balanced on volatility and relative spread but severely imbalanced on price and dollar volume, so the wide-spread controls do not provide common support. Only 17 of 75 control stocks fall inside the treated price range ($5.9–$177.8). A price-and-volatility-matched control is the student's reserved next step.

Table 11: The DiD under alternative screen definitions

Screen definition	DiD (bps)	t	SE	#stocks	N
Baseline (all 150 stocks)	+0.732***	(+4.18)	0.175	150	5820
Tighter treated screen (dqs_scr $\leq$ $0.012)	+0.820***	(+4.58)	0.179	136	5281
Narrower control screen (dqs_scr $\geq$ $0.03)	+0.916***	(+4.44)	0.207	131	5082
Donut: drop names near $0.015 cutoff	+1.003***	(+4.79)	0.210	117	4543
Drop mega-caps (screen price > $500)	+0.394***	(+2.59)	0.152	133	5157

Notes: DV = relative quoted spread (bps), two-way fixed-effects DiD (Treated×Post), SE clustered by stock, re-estimated on subsets of the same 150 stocks under the stated observable screen. The wrong-signed, significant estimate survives every reasonable redefinition, including dropping mega-caps. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 12: Randomization inference (500 permutations of treatment)

Outcome (DV)	Actual DiD	Perm. 95% interval	RI p-value
Relative spread (bps)	+0.7316	[-0.3701, +0.3568]	<0.002
Dollar spread (cents)	+4.3598	[-1.9935, +2.0250]	<0.002

Notes: The time-invariant treatment label is permuted across the 150 stocks 500 times and the two-way fixed-effects DiD is re-estimated each time. "Perm. 95% interval" is the 2.5–97.5 percentile of the permutation distribution; the RI p-value is the share of permutations with $|\text{coefficient}| \geq |\text{actual}|$. The actual estimate lies far outside the permutation distribution: because random assignment ignores price and liquidity, this locates the estimate in the size/price structure of *which* stocks are treated, not in the reform. See Figure 4.

Table 13: Minimum detectable effects at 80% power

Outcome (DV)	SE	MDE(80%)	Pre treated mean	MDE % of mean
Relative spread (bps)	0.1751	0.4906	3.555	13.8%
Dollar spread (cents)	0.9536	2.6716	1.334	200.2%
Eff.-spread proxy (bps)	0.3950	1.1067	6.464	17.1%

Notes: MDE(80%) = $(z_{0.975} + z_{0.80})\times$SE $\approx 2.80\times$SE of the preferred-specification DiD, expressed as a share of the pre-period treated mean. For the relative spread the MDE is 0.49 bps; a \$0.01→\$0.005 narrowing at the median treated midpoint (\$48.90) is 1.02 bps—above the MDE—so the reform's margin would be detectable *if* the daily close could represent a half-cent spread (Table 8 shows it never does). The dollar-spread MDE is large relative to its small treated mean and is read cautiously.

Figures

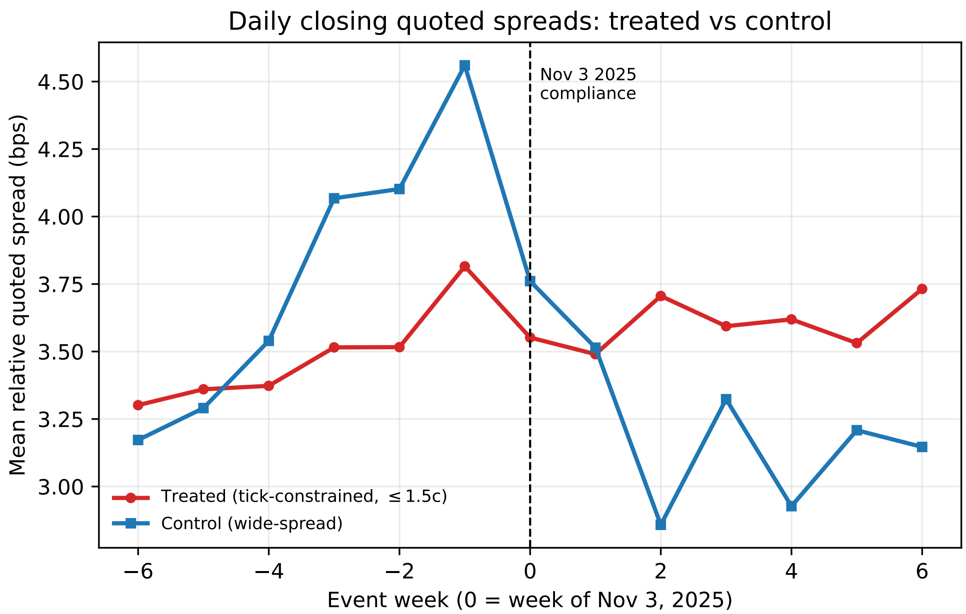

Figure 1: Event-time mean relative quoted spread (bps) for treated (tick-constrained) and control (wide-spread) stocks, by week relative to the Nov 3, 2025 compliance date. The two series do not move in parallel, and neither shows a clean step at week 0.

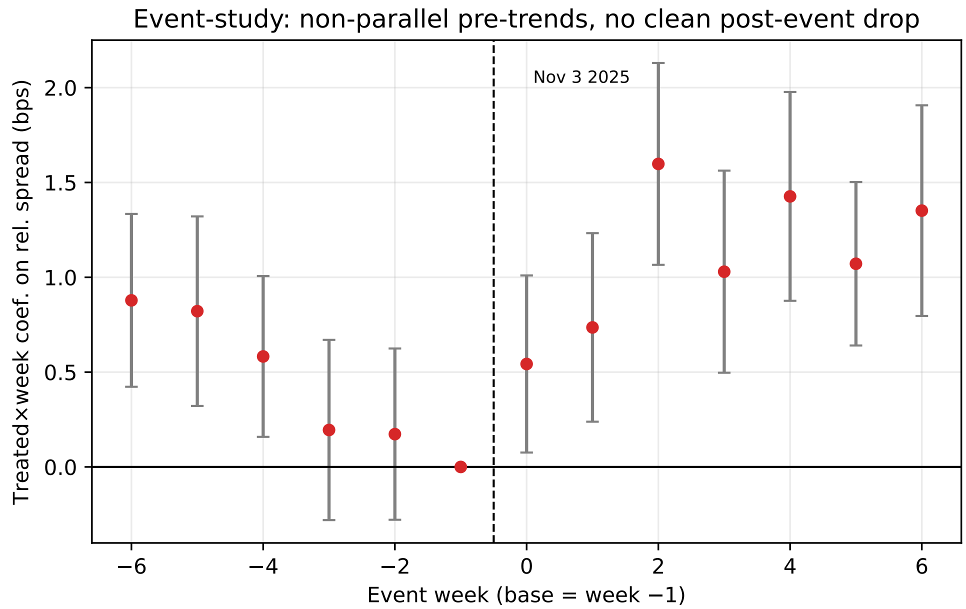

Figure 2: Event-study DiD coefficients (Treated×week) on the relative quoted spread, with 95% confidence intervals, base week = −1. Significant pre-event coefficients reveal non-parallel pre-trends; there is no downward step at the event.

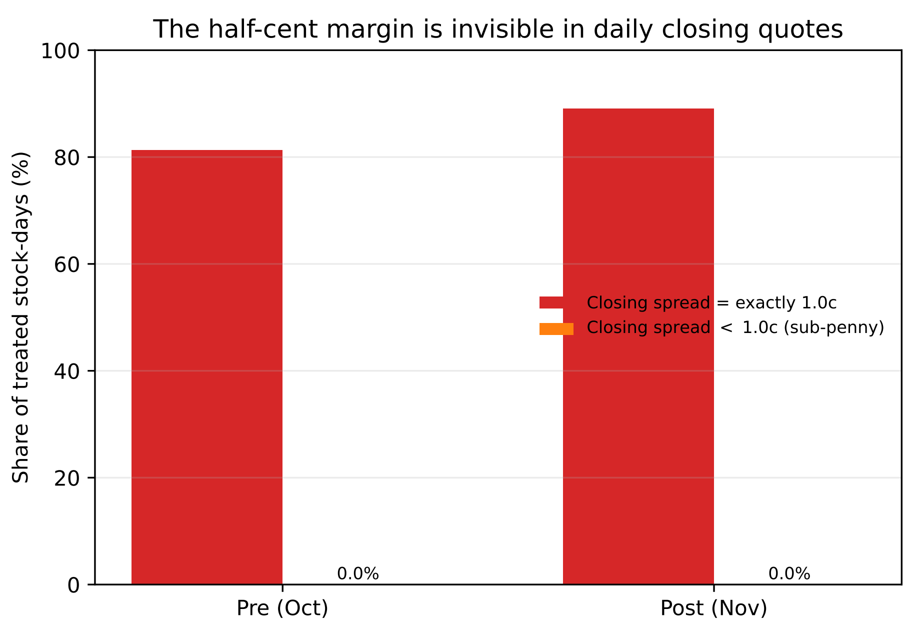

Figure 3: Penny-grid censoring. Among treated (tick-constrained) stocks, the share of stock-days with a sub-penny closing spread is 0.0% both before and after the reform, while the share pinned at exactly one cent rises. The half-cent margin is invisible in daily closing quotes.

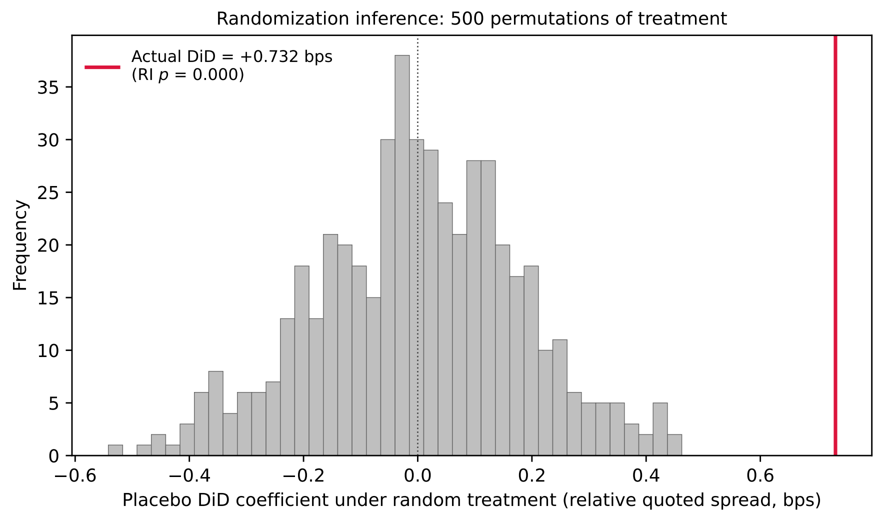

Figure 4: Randomization inference for the relative-quoted-spread DiD. Histogram: the distribution of the DiD coefficient under 500 random permutations of the treatment label across the 150 stocks. The actual estimate (red line, +0.73 bps) lies far in the right tail (RI $p < 0.002$); because random assignment ignores price and liquidity, this locates the estimate in the price structure of *which* stocks are treated, not in the November 3 reform.